

MELODY YIYUAN ZHANG

Ph.D. Candidate, University of Michigan, Ann Arbor, MI 48105

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EDUCATION

Ph.D. in Chemical Engineering and Scientific Computing	Expected <i>December 2026</i>
University of Michigan, Ann Arbor, MI	GPA: 3.81
B.S. in Chemical Engineering	<i>May 2022</i>
Purdue University, West Lafayette, IN	GPA: 3.82

AWARDS

Empowering Research with AI Award, AI Institutes at Michigan (AIIM)	<i>2026</i>
Best Overall Poster Award, U-M Annual Data Science & AI Summit	<i>2025</i>
Michigan Institute for Computational Discovery & Engineering (MICDE) Fellowship	<i>2025</i>
2nd Place, U Michigan Ross Business+Tech Datathon	<i>2025</i>
1st Place, College of Engineering Undergraduate Research Conference	<i>2021</i>
Research Grant, Office of Undergraduate Research, Purdue University	<i>2021</i>
Purdue Women in Engineering Program Scholarship	<i>2021</i>
Chemical Engineering Stephen Craig Outstanding Junior Award	<i>2021</i>
AstraZeneca Rising Star Award for initiative, innovation, and productivity	<i>2021</i>

RESEARCH EXPERIENCE

Graduate Student Research Assistant , Lindsey Group & Glotzer Group	<i>Oct. 2022 – Present</i>
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Machine-Learning Framework for Inverse Design of Polymer-Grafted Nanoparticles

- Developed a machine-learning interaction framework integrated with digital alchemy inverse design, enabling direct optimization of polymer-grafted nanoparticle physicochemical attributes toward target self-assembled structures.
- Designed an enhanced sampling strategy that concentrates training data in physically relevant regions of configuration space, improving sampling efficiency by approximately $10\times$ while retaining accuracy.

Orientation-Aware Machine-Learning Model for Anisotropic Nanoparticle Interactions

- Designed and implemented a generalized, orientation-dependent machine-learning interaction model for anisotropic nanoparticles, extending machine-learned interaction modeling to particles with non-spherical shapes and directional interactions.
- Established physics-informed training strategies that exploit known particle symmetries and adaptive refinement to focus sampling on symmetry-distinct, high-information configurations, reducing required training data by approximately $100\times$ while improving data quality and model quality.
- Enabled molecular dynamics simulations with the anisotropic interaction model, extending the framework beyond static energy prediction to dynamical simulations of nanoparticle self-assembly.

Machine-Learned Design Coordinates for Sequence-Defined Nanoparticles

- Developed a method that learns continuous coordinates for discrete ligand sequences together with a physics-based model of particle interactions, enabling inverse design directly on the fitted interaction model without a separate black-box predictor.
- Demonstrated that monomer sequence order, rather than composition alone, can strongly alter nanoparticle interactions, establishing ligand sequence as an independent design variable for controlling self-assembly.

Undergraduate Research Assistant , Dou Group, Purdue University	<i>Jan. 2020 – May 2022</i>
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- Synthesized ligand-assisted perovskites for defect passivation and phase stabilization, achieving photovoltaic devices with 20.1% power conversion efficiency.
- Streamlined photovoltaic device data analysis for rapid evaluation of performance and ambient stability.

RESEARCH PUBLICATIONS

Zhang, M. Y., Lee, S. K. A., Glotzer, S. C., and Lindsey, R. K. Aniso-ChIMES: Exact Symmetry- and Shape-Adapted Machine-Learned Interaction Models for Anisotropic Nanoparticles. Manuscript in preparation.

Zhang, M. Y., Glotzer, S. C., and Lindsey, R. K. From Spheres to Anisotropy in Machine Learning Potentials for Nanomaterials. Review article in preparation for *MRS Bulletin*.

Zhang, M. Y.*, Lee, S. K. A.*, Glotzer, S. C., and Lindsey, R. K. A Generalized Machine-Learning Framework for Developing Alchemical Many-Body Interaction Models for Polymer-Grafted Nanoparticles. *J. Chem. Theory Comput.* 2025, 21, 9853–9867. DOI: [10.1021/acs.jctc.5c00901](https://doi.org/10.1021/acs.jctc.5c00901) (*equal contribution)

Finkenauer, B., **Zhang, Y.**, Ma, K., Turnley, J. W., Schulz, J., et al. Amine-Thiol/Selenol Chemistry for Efficient and Stable Perovskite Solar Cells. *J. Phys. Chem. C* 2023, 127, 930–938. DOI: [10.1021/acs.jpcc.2c06967](https://doi.org/10.1021/acs.jpcc.2c06967)

Zhang, Y. Interface Engineering to Kinetically Trap the Photovoltaic Crystal Phase of Formamidinium Lead Iodide Perovskite. *Journal of Purdue Undergraduate Research* 2021, 11, Article 15. DOI: [10.7771/2158-4052.1533](https://doi.org/10.7771/2158-4052.1533)

SCIENCE WRITING & COMMUNICATION

Materials News Writer, *MRS Bulletin*

June 2026 - Present

- Mazalov, I., **Zhang, M. Y.** Turning industrial nanoparticle furnace into a high-entropy alloy factory. *MRS Bulletin* (2026). DOI: [10.1557/s43577-026-01205-9](https://doi.org/10.1557/s43577-026-01205-9).
- **Zhang, M. Y.**, Mazalov, I. Graph theory reveals how disorder enhances optical response in nanoparticle assemblies. *MRS Bulletin* (2026). DOI: [10.1557/s43577-026-01171-2](https://doi.org/10.1557/s43577-026-01171-2).
- Mazalov, I., **Zhang, M. Y.** Shape-shifting nanoparticles introduce dynamic path for CO₂ conversion. *MRS Bulletin* (2026). DOI: [10.1557/s43577-026-01170-3](https://doi.org/10.1557/s43577-026-01170-3).

Meeting Scene Reporter, Materials Research Society (MRS)

April–May 2026

- **AI-Driven Integration of Molecular Modeling and Experimental Data.** *MRS Meeting Scene*, Symposium MT04, 2026.
- **Accelerating Battery Materials Discovery with AI- and Physics-Powered Molecular Universe Platform.** *MRS Meeting Scene*, Symposium MT02, 2026.
- **Solvation Structure Modulation in Aqueous Electrolytes: Stabilizing Pyrazine Radicals and Facilitating Electrochemical Nitrate Reduction.** *MRS Meeting Scene*, Symposium MT04, 2026.
- **Bespoke Machine-Learned Interatomic Potentials – Pushing the Limits of Accuracy and Efficiency.** *MRS Meeting Scene*, Symposium MT03, 2026.
- **Optimization Method of Monomer Composites’ Ratio Leading to the Best Polymer Property by Using Machine Learning Method and Quasi-Quantum Computing Method.** *MRS Meeting Scene*, Symposium MT04, 2026.

CONFERENCE PRESENTATIONS

Zhang, M. Y., Lee, S. K. A., Glotzer, S. C., and Lindsey, R. K. A Generalized Alchemical Machine-Learned Interaction Potential for Modeling Anisotropic Nanomaterial Interactions. Oral presentation, *Material Research Society*, Spring 2026, Honolulu, HI.

Zhang, M. Y., Lee, S. K. A., Glotzer, S. C., and Lindsey, R. K. A Generalized Framework for Developing Alchemical Coarse-Grained Interaction Models for Self-Assembling Polymer-Grafted Nanoparticles. Oral presentation, *Midwest Thermodynamics and Statistical Mechanics Conference*, Summer 2025, Madison, WI.

Zhang, M. Y., Lee, S. K. A., Laubach, B., Glotzer, S. C., and Lindsey, R. K. Generating Nanomaterial Design Surfaces with Physics-Informed Machine Learning Models. Oral presentation, *American Institute of Chemical Engineers*, Fall 2024, San Diego, CA.

Zhang, M. Y., Lee, S. K. A., Glotzer, S. C., and Lindsey, R. K. Towards Mapping Nanomaterial Design Space with Physics-Informed Machine-Learned Models. Poster presentation, *Midwest Thermodynamics and Statistical Mechanics Conference*, Summer 2024, Ann Arbor, MI.

Zhang, M. Y., Lee, S. K. A., Laubach, B., Tsai, S., Anderson, J. A., Glotzer, S. C., and Lindsey, R. K. A Machine-Learning Strategy for the Physics-Informed Design of Nanomaterials Using ChIMES. Poster presentation, *American Physical Society*, Spring 2024, Minneapolis, MN.

PROFESSIONAL EXPERIENCE

Advanced Modeling and AI Solutions Intern, Micron Technology, Boise, ID

May – Aug. 2026

- Developed a material search agent that standardizes naming across multiple databases and uses fuzzy matching algorithms to interpret user queries, enabling accurate retrieval for automation workflows.
- Built an end-to-end workflow integrating material-property retrieval with data-driven process modeling and automated Silvaco TCAD structure generation to demonstrate correlation between material properties and device specifications.

Graduate Student Instructor, Mathematical Analysis for Chemical Engineers *Aug. – Dec. 2025*

- Taught and supported a graduate-level applied mathematics course covering linear algebra, ODEs/PDEs, numerical methods, statistics, and optimization for chemical engineering applications.
- Led office hours and problem-solving sessions, developed solution materials, and provided individual guidance on analytical and computational approaches to advanced engineering problems.
- Supported homework and exam preparation and consistently received positive student feedback for clear, effective instruction in a rigorous and method-intensive course.

Validation Engineering Intern, AstraZeneca, Mt. Vernon, IN *May – July 2021*

- Analyzed product quality data for two high-volume active pharmaceutical ingredients using BIOVIA Discoverant to support annual quality reporting and process validation.
- Designed and implemented an interactive Tulip application to streamline equipment and process risk assessments for engineering and operations teams.

Resident Assistant, Purdue University *Nov. 2020 – May 2022*

LEADERSHIP EXPERIENCE

Liaison to the Director, Center for Simulation and Data Intensive Research in Chemical Engineering (CSiDIR-CHE), University of Michigan *2023 – 2024*

Engineering Peer Mentor, Women in Engineering Program, Purdue University *2020*

Engineering Peer Mentor, Boiler Student Network, Purdue University *2020*

Events Planning Chair, UR Multicultural Connections, Purdue University *2020 – 2021*

Purdue AIChE Officer, Student Relations & Faculty Relations Committee *2020*

Vice President, Wiley Resident Hall Club, Purdue University *Aug. 2019 – 2020*

RELEVANT COURSEWORK

Design & Analysis of Algorithms	Machine Learning	Parallel Computing
Applied Math for Chemical Engineering	Numerical Linear Algebra	Big Data Management Tools
Statistical & Irreversible Thermodynamics	Simulation of Condensed Matter	

SKILLS

Programming & HPC: Python, C++, Bash, SQL, Git, Parallel Computing (MPI, OpenMP)

Machine Learning: PyTorch, scikit-learn, VAE, GNN, RAG, Agentic workflow, Database construction

Simulation & Visualization: Silvaco (VPA, TCAD), LAMMPS, HOOMD-blue, OVITO, VMD, Blender